Theory of Computation-PROBLEM SET 4

May 6, 2026

Due May 21st. Upload to learn.tsinghua.edu.cn.

Problem 1

Let L be a language and A be a randomized polynomial-time algorithm such that, for any input x, if $x \in L$,

$$Pr[A \text{ accepts } x] \geq 2/|x|$$

if $x \notin L$,

$$Pr[A \text{ accepts } x] \leq 1/|x|$$

Prove that $L \in \mathsf{BPP}$. (Recall that we defined BPP as the set of languages that have a randomized polynomial-time algorithm with the error probability $\leq 1/3$. Here $A(x)$ is equivalent to $M(x, y)$ for random string y as shown in class.)

Problem 2

Suppose there is a polynomial-time algorithm to decide whether a given Boolean formula $\phi(x_1, \cdots, x_n)$ has a satisfying truth assignment. (In other words, suppose $\mathsf{P} = \mathsf{NP}$.) Show that this implies that we can actually find a satisfying assignment for any Boolean formula ϕ in polynomial time, whenever one exists. [Interestingly, integer factoring does not satisfy this property.]

Problem 3

Show that $\mathsf{P} \neq \mathsf{SPACE}(n)$.

Problem 4

Show that if $\mathsf{NP} \subseteq \mathsf{BPP}$, then $\mathsf{NP} \subseteq \mathsf{RP}$, hence $\mathsf{NP} = \mathsf{RP}$.

Problem 5

A *directed acyclic graph (DAG)* is a finite directed graph with no directed cycles. Suppose a directed simple graph G of n vertices (with no multi-edges and self-loops and $n \geq 3$) is given by its adjacency matrix with $n(n-1)$ elements. What is the certificate complexity of DAG? Prove your answer. (You need to consider both 0-certificates and 1-certificates.)

Problem 6 (Optional)

In class we define the MA class by perfect completeness. Show that it is the same class if we do not have perfect completeness. (That is, $L \in \mathsf{MA}$ iff there is a polynomial time language R, such that if $x \in L$, $(\exists m) Pr_r[(x, m, r) \in R] \geq 2/3$, and if $x \notin L$, $(\forall m) Pr_r[(x, m, r) \in R] \leq 1/3$.)